CHAPTER 15

MAKING LOAN AND INTEREST CALCULATIONS

If you're involved in calculating percentages and equations, you'll find the WordStar print-time math capabilities helpful. This chapter shows how to figure the amount of interest a customer will pay on a large purchase, and what the monthly payments will be. The example is set up as a merge print document so the letter can go out to a group of customers.

SETTING UP THE MERGE PRINTING PARAMETERS

You start the document by giving the commands to set up the page design.

1. Open a new document file.

2. Choose Layout→Page Numbering (Alt+L,N) to display the Page Numbering dialog box. Clear the Print Page Numbers check box. Press F10 or choose OK.

   WordStar inserts the dot command **.op** to omit page numbers, since this is the only page of the letter.

3 Choose Utilities→Merge Print Commands→Data File (Alt+U,P,D). Type the name of the data file. Press F10 or choose OK.

WordStar inserts the **.df** command.

4 Choose Utilities→Merge Print Commands→Name Variables (Alt+U,P,N). Type the name of each variable in your data file; separate the variable with a comma. Do not type ampersands in the dialog box.

The data file must include variables for the amount of the purchase or loan, the interest rate being charged, and the total number of payments (the length of the loan). In this example, the variables are

```
&purch&      amount of purchase
&int&        interest rate
&numpay&     number of payments
```

WordStar inserts a **.rv** command for the variables.

CALCULATING MONTHLY PAYMENTS

When you know the amount being financed, the interest rate, and the terms of purchase, you can use the following equation to determine the monthly payment.

$$P = A * i * (i + 1)^{n}/((i + 1)^{n-1})$$

In the equation

P is the amount of each payment.

A is the amount being financed.

i is the interest rate. To find the interest per payment, **i** is divided by the number of payments per year and must be converted from a percentage to a decimal.

n is the total number of payments. For example, monthly payments for 14 years total 168 payments.

Follow these steps to have WordStar calculate the payment amount.

1 Choose Utilities→Merge Print Commands→Set Variable to Math Result (Alt+U,P,V). In the Variable Name box, type **i**. In the Mathematical Expression box, type **&int&/1200**. Press F10 or choose OK.

WordStar inserts the following command into your file:

```
.ma i=&int&/1200
```

This command converts the interest rate from a percentage amount to a decimal figure.

2 Choose Utilities→Merge Print Commands→Set Variable to Math Result (Alt+U,P,V). In the Variable Name box, type **P**. In the Mathematical Expression to Be Calculated box, type this:

```
&purch& * &i& * ((&i&+1)^&numpay&)/(((&i&+1)^&numpay&)-1)
```

Press F10 or choose OK.

WordStar inserts the following command into your file:

```
.ma P=&purch& * &i& * ((&i&+1)^&numpay&)/(((&i&+1)^&numpay&)-1)
```

This command determines the amount of the monthly payment.

SETTING VARIABLE FORMATS

Next you'll determine the format for the variables that contain numeric values.

1 Choose Utilities→Merge Print Commands→Set Variable (Alt+U,P,S).

2 In the Variable box, type **a**.

3 In the Data box, type **$$$,$$$.99**. Press F10 or choose OK.

The following command is inserted into your file:

```
.sv a=$$$,$$$.99
```

WordStar will put a dollar sign in front of the number and a comma after the thousands place.

Next, you'll tell WordStar which variables to print with the dollar sign format.

1 Choose Utilities→Merge Print Commands→Set Variable (Alt+U,P,S). In the Variable box, type **purch**. In the Data box, type **&purch/a&**. Press F10 or choose OK.

WordStar inserts the following command into your file:

```
.sv purch=&purch/a&
```

2 Choose Utilities→Merge Print Commands→Set Variable (Alt+U,P,S). In the Variable box, type **P**. In the Data box, type **&P/a&**. Press F10 or choose OK.

WordStar inserts the following command into your file:

```
.sv P=&P/a&
```

TYPING YOUR LETTER

After you define the variables and create the equations, type the letter to your customers.

Start by typing the name, address, and salutation, using the variable names in the **.rv** commands at the beginning of the file.

Note: If there is no data in certain fields, use the /o option to avoid blank lines.

1 Type this:

```
&Mr-Ms& &name&
&addr1/o&
&addr2/o&
&city&, &state& &zip&

Dear &Mr-Ms& &name&,
```

2 Type the text for the letter. Use the new &P& and &i& variables you created with the **.ma** commands.

```
Recently, you purchased &item& from our company.  The
total purchase price of &item& was &purch&.  This letter
is to confirm the interest rate and payment schedule for
your purchase.

The interest rate for your purchase is &i&%.  This works
out to &numpay& payments of &P& each.

If you have any questions, please contact me at our main
office.

Sincerely,


Mr. Perry, Credit Manager
```

3 Choose Insert→Page Break (Alt+I,P) to insert a page break.

WordStar inserts **.pa** into the document.

Your letter looks similar to this.

```
.op
.df finance.dta
.rv Mr-Ms,name,addr1,addr2,city,state,zip
.rv item,purch,int,numpay
.ma i=&int&/1200
.ma P=&purch&*&i&*((&i&+1)^&numpay&)/(((&i&+1)^&numpay&)-1)
.sv a=$$$,$$$.99
.sv purch=&purch/a&
.sv P=&P/a&
&Mr-Ms& &name&
&addr1/o&
&addr2/o&
&city&, &state& &zip&

Dear &Mr-Ms& &name&,

Recently, you purchased &item& from our company.  The
total purchase price of &item& was &purch&.  This letter
is to confirm the interest rate and payment schedule for
your purchase.
```

```
The interest rate for your purchase is &i&%.  This works
out to &numpay& payments of &P& each.

If you have any questions, please contact me at our main
office.

Sincerely,


Mr. Perry, Credit Manager
.pa
```

When you print this form letter, WordStar calculates the monthly payments and interest rates for all the customers in your data file.